

Common container patterns for AI/ML deployment

Microservices architecture: Many AI/ML applications are composed of various components, such as data preprocessing, model training, and inference services. Kubernetes supports a microservices architecture by enabling each component to run in its own container and communicate with others via APIs. This approach boosts scalability and maintainability.

Job scheduling: Training AI/ML models can be time consuming and resource-intensive. Kubernetes jobs and cron jobs allow teams to schedule and manage these training tasks efficiently. Teams can define jobs that run once or on a schedule, ensuring that training processes can be automated and monitored.

Model serving: Once trained, AI/ML models need to be served for inference. Kubernetes supports scalable model serving through tools like TensorFlow Serving, KFServing, and Seldon Core. These tools integrate seamlessly with Kubernetes, providing high-performance and scalable inference services.

Data pipelines: Kubernetes enables the orchestration of end-to-end data pipelines, from data ingestion to preprocessing, training, and deployment. Tools like KubeFlow and Apache Airflow, which run on Kubernetes, provide robust platforms for managing these complex workflows.

Deploying generative AI applications with Docker and Kubernetes

To deploy a generative AI application using Docker and Kubernetes, follow these steps.

Containerise the application: Use Docker to create a container image of your generative AI application. This involves writing a Dockerfile that specifies the application's dependencies and configuration.

Push the image to a registry: Push the Docker image to a container registry,

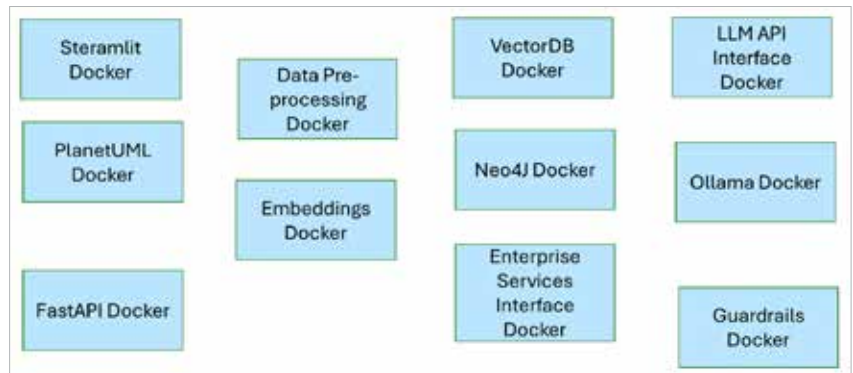


Figure 2: Containers for building generative AI applications

such as Docker Hub or a private registry, making it accessible to Kubernetes.

Create Kubernetes manifests: Write Kubernetes manifests (YAML files) that define the desired state of your application, including deployments, services, and configurations.

Deploy to Kubernetes: Use kubectl commands to deploy the application to a Kubernetes cluster. Kubernetes will handle the orchestration, ensuring the application runs as specified.

Monitor and scale: Use Kubernetes' monitoring and scaling features to ensure

the application performs optimally. Adjust the number of replicas and resources as needed to meet demand.

By leveraging the right tools and frameworks, developers can create scalable, efficient, and secure generative applications that deliver significant value across various domains. Developers can use containers, Docker and Kubernetes to streamline the deployment process, reduce operational complexity, and ensure their applications run efficiently in any environment. **END** 🐧

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